

Prolongation, projection, and the quantifier discipline: an algorithm for the ansatz-plus-PDE solution space and the resolution of the $v = v_1x + v_2y + v_3z$ puzzle

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Abstract

A companion note to *A New Method of Solving PDEs*. Given an ansatz $A(c)$ (a differentially triangular family with constant parameters c) and a PDE P , we survey the algorithm families available for computing the solution space of the combined problem, then develop a proposed three-phase algorithm: prolong $A \cup \{P\}$ to differential closure while tracking the inequations actually used (Phase I), recurse Thomas-style on the tracked inequation loci (Phase II), and project each resulting system onto the constants under the quantifier discipline *independent variables universal, dependent variables and constants existential* (Phase III). The intellectual center of the note is a puzzle: applied naively, the universal coefficient-zeroing of Phase III would read the ansatz-defining equation $v = v_1x + v_2y + v_3z$ as forcing $v_1 = v_2 = v_3 = 0$. We resolve the puzzle by making the quantifier discipline precise. Universal quantification is implemented by *coefficient collection* and is sound only over symbols that are algebraically free of one another — which the fully reduced remainder’s variables are, and a defining equation’s leader is not; existential quantification is implemented by *Gröbner elimination* and is sound only *after* differential closure (prolongation to passivity). The naive Phase III interleaves the two operations in an order in which they do not commute, and we exhibit small examples where it over-constrains (the v -equation) and where it under-constrains (returning $\mathbb{C}$ where the correct existence locus is $\{c = 1\}$). Corrected, the algorithm computes the per-cell *existence* locus — the joint-variety projection $\mathcal{V} \cup \{\text{partial strata}\}$ — with the *membership* locus $\mathcal{V}$ selectable per cell by flipping the quantifier on the parametric derivatives, equivalently by the dimension-preservation criterion. A final section takes up the route that actually runs on hydrogen: decompose the *ansatz alone*, then bring in the PDE cell by cell. We locate its blind spot exactly (it loses a stratum iff the reduced remainder carries a parametric jet), record why re-completing $T_j \cup \{P\}$ per cell exhausts memory on five of the 29 cells, and propose the *fixed-chain prolongation*: leave the passive cell untouched, prolong the remainder against it, and test stabilization by D -closure of a commutative ideal — no Δ -polynomials, no splitting, and a variable set that cannot grow.

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1 Problem statement: two loci, not one

Let $A(c) = \{a_1, \dots, a_k\}$ be the *ansatz*: a differentially triangular set of differential polynomials in dependent variables $\Psi, v, \dots$ over independent variables x, y, z (derivations $\partial_x, \partial_y, \partial_z$), whose coefficients involve constant parameters $c \in \mathbb{C}^n$. Let P be the PDE. The running concrete case is the hydrogen ansatz of the paper [1]:

$$\begin{aligned}
& (a_0 + a_1v) \Psi'' + (b_0 + b_1v) \Psi' + (c_0 + c_1v) \Psi = 0, \\
& v - (v_1x + v_2y + v_3z + v_4r) = 0, \quad r^2 - (x^2 + y^2 + z^2) = 0, \\
& \Psi_x - \Psi'v_x = 0, \quad \Psi_y - \Psi'v_y = 0, \quad \Psi_z - \Psi'v_z = 0,
\end{aligned} \tag{1}$$

with P the Schrödinger equation $-\frac{1}{2}(\Psi_{xx} + \Psi_{yy} + \Psi_{zz}) - \frac{1}{r}\Psi = E\Psi$ and eleven constants $c = (E, v_1, \dots, v_4, a_0, a_1, b_0, b_1, c_0, c_1)$.

“The solution space of ansatz + PDE” admits two readings, and the entire subject of this note turns on keeping them distinct [2]:

$$\mathcal{M} = \{c : P \in [A(c)]\} \quad (\text{membership locus: every ansatz member solves } P), \quad (2)$$

$$\mathcal{E} = \pi_c(\text{Sol}(A) \cap \text{Sol}(P)) \setminus \{\text{trivial}\} \quad (\text{existence locus: some nonzero member solves } P). \quad (3)$$

Here $[A(c)]$ is the differential ideal generated by the ansatz at c , and “trivial” excludes the identically-zero solution, which for the linear-homogeneous class solves everything at every c and must be quotiented away before either locus is informative [3, §2bis, “Two saturations”].

$\mathcal{M}$ is a *universal* condition over the ansatz family; $\mathcal{E}$ is *existential*. Always $\mathcal{M} \subseteq \mathcal{E}$, and the difference

$$\mathcal{E} \setminus \mathcal{M} = \{\text{partial-solution strata}\}$$

consists of parameter values where a *proper nontrivial subfamily* of the ansatz solves P while the family as a whole does not [2, Def. 1]. The paper’s Algorithm 1 (reduce P modulo A , project the remainder onto the constants) computes $\mathcal{M}$; decomposing the combined system $A \cup \{P\}$ and projecting computes $\mathcal{E}$. The two-line example

$$A : \Psi'' - c\Psi, \quad P : \Psi' - \Psi$$

separates them: $\mathcal{M} = \emptyset$ (no c makes the whole two-dimensional family $K_1 e^{\omega x} + K_2 e^{-\omega x}$ obey $\Psi' = \Psi$), yet $\mathcal{E} = \{c = 1\}$ (the subfamily $K e^x$ solves at $c = 1$) [2, Ex. 1].

The paper’s own motivation for treating A and P *on an equal footing* — one combined decomposition rather than reduce-then-project — is precisely that the partial strata are invisible to the membership formulation by construction, and that some of them carry physically meaningful solutions: in the hydrogen differential Thomas run, the bad-locus cell at $E = -1/8$ is the hydrogen $2s$ state $(1 - r/2)e^{-r/2}$, whose constants provably fail the projection system of Algorithm 1 [6]. Any algorithm that adjoins P to A must therefore be judged against *both* targets: which locus does it compute, and can it be tuned to compute the other?

2 Survey: the algorithm families

Four routes to the ansatz-plus-PDE solution space are on the table. The conceptual comparison — what question each answers and what triggers a case split — is developed at length in [4]; we summarize, adding only what this note needs. Prior-art verdicts and full citations are in the litsearch report [5].

2.1 Reduce-then-project (the paper’s Algorithm 1)

Ritt’s full reduction of P modulo A over the field $\mathbb{Q}(c)$ produces $h \cdot P - r \in [A]$ with r fully reduced and h a power product of initials and separants of A (the “denominator”). The projection step treats each surviving non-constant symbol as an independent indeterminate, collects

$$r = \sum_{\alpha=1}^N m_{\alpha} p_{\alpha}(c), \quad (4)$$

m_{α} distinct power products in the non-constants, and returns $\mathcal{V} = V(p_1, \dots, p_N)$ as minimal primes (commutative GTZ). For hydrogen the collection runs over exactly x, y, z, r, Ψ, Ψ' [1, §Projection Step], yields 28 constant equations, and the five minimal primes reproduced by `joca.sage`.

Question answered: the membership locus, $\mathcal{V} = \mathcal{M}$ (on the locus $h \neq 0$; Theorem 1’s completeness hypothesis). *Splits:* none in the good case — one reduction, one GTZ call; the general Algorithm 2 recurses only on the *bad locus* where the hypotheses of the completeness theorem fail. *Cost/faithfulness:* the constants never enter the differential engine (they live in the coefficient field), which is the practical lever — and the price is that the bad-locus strata (e.g. $a_0 = a_1 = 0$, carrying the $1s$ and $2s$ states as first-order-ODE solutions) are outside the guarantee.

The closest published relative is Chaharbashloo–Basiri–Rahmany–Zarrinkamar [23]: substitute an *explicit* physics ansatz $\psi = g(r)e^{y(r)}$ into the wave equation, equate powers of r by hand, and solve the resulting constant system by a commutative Gröbner basis. That is exactly a reduce-then-project computation — and, as §5 will make precise, the reason their raw power-equating is sound is that an explicit ansatz has no defining equations left: substitution has already eliminated every dependent leader.

2.2 Differential Thomas decomposition

Bächler–Gerdt–Lange–Hegermann–Robertz [11] (implementation for PDE systems: the MAPLE package TDDS [12]; project Sage port in progress, `joca-thomas-native-dt.sage` / `thomas_cells.py`). Decomposes any system into finitely many *disjoint* simple systems (cells): each differentially triangular, *passive* (Janet-involutive: every non-multiplicative prolongation reduces to zero), with attached inequations (initials, separants, squarefreeness witnesses).

Question answered: the complete, disjoint stratification of $\text{Sol}(A \cup \{P\})$ — the existential object; the constant-space shadow of the cells is $\mathcal{E}$ together with its case structure. With the constants modeled as $\dot{c}_i = 0$, the constant-only inequations become parameter-degeneracy conditions and the decomposition is automatically comprehensive over the constants. *Splits:* wherever the *description* degenerates — every initial, separant, or discriminant of every equation processed during completion, whether or not its vanishing changes anything about existence. *Status here:* the decomposition of the *ansatz alone* completes with 29 cells; the decomposition of the *combined* system $A \cup \{P\}$ does not complete, and neither does its per-cell staged form on five of the 29 cells (§7). The $E = -1/2$ and $E = -1/8$ strata surface on ansatz cells 26/27 (`zero {a0,a1}`, `zero {a0,a1,b1}`) not as splits of the decomposition — E occurs in P and nowhere in A , so no decomposition of A can split on it — but as output of the projection run on those cells [6, 7].

2.3 Parametric Rosenfeld–Gröbner

Fakouri–Rahmany–Basiri [15]; the non-parametric engine is Boulier’s BLAD/BMI implementing Rosenfeld–Gröbner [14]. Regular differential chains represent $[C] : H_{\mathcal{C}}^{\infty}$; a comprehensive-Gröbner-style bookkeeping splits parameter space wherever an initial or separant’s vanishing is ambiguous mid-elimination.

Question answered: per parameter cell, a regular-chain description of the (saturated) solutions — existential in character, but of the *saturation* $[C] : H_{\mathcal{C}}^{\infty}$, so strata inside initial/separant loci are systematically divided out rather than kept. *Splits:* parameter conditions arising during elimination. *Status here:* the project port (`~/parametric-rg`) reproduces the paper’s examples; on hydrogen the parameter branching is cheap but the differential reduction hits the BLAD coefficient-swell wall [4]. Saturation-unfaithfulness is not hypothetical: reducing the PDE against the RG chain’s bare equations re-admits the spurious prime (a_0, a_1) while absorbing (losing) the two genuine strata inside it [10, `joca-rg.sage` discussion].

2.4 Bounded prolongation to the involutive order, then pure commutative algebra

The route developed in [3]: prolong $A \cup \{P\}$ to an order N at which the system is involutive, treat every jet of order $\leq N$ as an independent algebraic indeterminate, form the *un-saturated* polynomial ideal $J = ((A \cup \{P\})^{[N]}) \subseteq K[\text{jets}_{\leq N}, x, y, z, r, c]$, and finish with any commutative-algebra algorithm. The differential engine is touched exactly once. The semantic core is published (Lange-Hegermann’s Theorem 4.1 [13]: truncated solution sets of an involutive system equal the solution sets of its algebraized truncation; the construction itself appears inside Golubitsky–Kondratieva–Ovchinnikov’s treatment of the generalized Ritt problem [17], there with the bad-locus-deleting saturation $: H_{\mathcal{C}}^{\infty}$ that this route deliberately omits). Comprehensive analogues on the algebraic side: comprehensive involutive systems [18] and comprehensive triangular decomposition [19, 20].

Question answered: the existence locus, $V((J : \Psi^{\infty}) \cap K[c]) = \overline{\mathcal{E}}$ — the $: \Psi^{\infty}$ quotient (delete only the zero section) is mandatory, and is a different operation from the H_A -saturation the route avoids [3, §2bis]. *Splits:* optional — a constructible (algebraic-Thomas) split on initials/separants/discriminants recovers the disjoint strata; the single-ideal route returns the Zariski closure, a superset that never loses a component. *Two corrections from the July 2026 implementation attempt* [6], both inherited by the algorithm of §3:

- The involutive order of the *ansatz alone* ($N_0(A) = 2$ for hydrogen, from the single- v structure) is **not** the involutive order of the *combined* system: the PDE–ODE critical pair (shared leader Ψ'') feeds conditions $D^k \rho$ (derivatives of the Ritt remainder) into the ideal only at order $2 + k$. No a-priori $N_0(A \cup \{P\})$ is known; *prolong-until-stable* — watch the elimination ideal $(J^{(N)} : \Psi^{\infty}) \cap K[c]$ ascend and stop at stabilization (Hilbert basis theorem) — is the algorithm, not a safety check.
- The direct implementation over all eleven constants as ring variables hits a comprehensive-elimination cliff (no completion even over $\text{GF}(32003)$ at $N = 2$); per-cell staging over the already-computed Thomas cells, or pointwise Nullstellensatz probes, are the practical fallbacks.

2.5 Summary table

	question answered	what triggers a split	principal limitation
Algorithm 1 (reduce+project)	membership $\mathcal{M}$ on $\{h \neq 0\}$	only bad-locus recursion (Alg. 2)	misses partial strata and the $h = 0$ strata by construction
differential Thomas	full disjoint stratification of $\text{Sol}(A \cup \{P\})$; shadow = $\mathcal{E}$	every initial / separant / discriminant met during completion	splitting cost; constants ride through the differential engine
parametric RG	per-cell regular chains of the <i>saturated</i> ideal	parameter ambiguity mid-elimination	saturation deletes in-locus strata; coefficient swell
bounded prolongation	$\overline{\mathcal{E}}$ (closure), strata optional	optional constructible split	no a-priori $N_0(A \cup \{P\})$; parametric elimination cliff
staged: Thomas on A , then P per cell (§7)	$\mathcal{M}$ per cell, <i>on every cell</i> — bad locus included	degeneracies of A ’s description only; P never enters the differential engine	blind to parametric-leader strata (Prop. 1); re-completing $T_j \cup \{P\}$ blows up on 5 of the 29 cells

3 The proposed algorithm

We now state the three-phase proposal this note is asked to develop, first as proposed, then — after the analysis of §§4–5 — in corrected form (§6).

3.1 Phase I: prolong, reduce, track

Prolong $A \cup \{P\}$ and auto-reduce until differential closure — every integrability condition (non-multiplicative prolongation, equivalently Δ -polynomial) reduces to zero. By the first correction of §2.4, this must be run as a completion loop (prolong $\rightarrow$ reduce $\rightarrow$ adjoin any nonzero residue, terminating by Ritt–Raudenbush), not as a single prolongation to a precomputed bound: for the combined system the constants-cutting conditions are the reduced derivatives $D^k \rho$ of the Ritt remainder, and no a-priori bound on k is available.

During every reduction step, record which reductor was used and how: *algebraic* steps divide by the reductor’s *initial*; *differential* steps (reduction against a proper prolongation θa) divide by its *separant*. In the algebraized picture the two cases unify: the initial of every proper prolongation θa is the separant of a , so the tracked set is simply “the initials of the (possibly prolonged) equations actually used.” The product of the tracked divisors is exactly Ritt’s denominator: the reduction identity is

$$h \cdot p \equiv r \pmod{(T^{[N]}),} \quad h = \prod_i I_i^{d_i} \prod_j S_j^{e_j}, \quad (5)$$

a polynomial identity, valid whether or not h vanishes; only the *conclusions* drawn from r (e.g. “ $r = 0$ hence p is a consequence”) require $h \neq 0$.

3.2 Phase II: recurse on the tracked inequations

For each tracked inequation h_i , split:

$$\text{Sol}(\Sigma) = \text{Sol}(\Sigma \cup \{h_i \neq 0\}) \sqcup \text{Sol}(\Sigma \cup \{h_i = 0\}),$$

keep $h_i \neq 0$ on the main branch, and recurse Phase I on the branch with h_i adjoined as an equation. Termination: adjoining h_i strictly enlarges the (radical differential) ideal, and strictly ascending chains of radical differential ideals terminate (Ritt–Raudenbush); when $h_i \in K[c]$ the split strictly descends the parameter dimension, the same descent that terminates Algorithm 2’s bad-locus recursion and CTD’s discriminant recursion. The output is a finite tree of systems ($E_i = 0$, $NE_i \neq 0$), each differentially closed (passive) relative to its inequations.

This is the Thomas/RG splitting discipline with lazy bookkeeping: Thomas also splits only on initials, separants, and squarefreeness conditions of equations it actually processes — not on arbitrary potential zero-divisors — so Phases I–II are best understood as a differential Thomas decomposition whose split set is generated *on demand* by the reduction trace. Two honest caveats. First, “used” must include the separants that license the *parametric-data freeness* of the final cell (each solved-form equation’s initial must be a unit for the Riquier existence theorem to apply); these are the separants of the final triangular set, which the completion loop does use, but an implementation that tracked only mid-reduction divisors and not final-form initials would emit cells on which Phase III’s existence semantics silently fail. Second, disjointness of the output requires carrying the complementary equations honestly ($h_i = 0$ branches), exactly as Thomas does; dropping them recovers Algorithm 1’s “main branch plus bad locus” shape.

3.3 Phase III as proposed — and why it cannot stand as stated

The proposal: in each equation of each leaf system, *group like terms by the independent variables and their algebraic extensions* (x, y, z, r with $r^2 = x^2 + y^2 + z^2$); because the equation must hold for all real values of the independents, set every grouped coefficient to zero; each coefficient involves constants and dependent variables; since a dependent variable need only take *some* value, *eliminate* the dependent variables from the coefficient system by Gröbner elimination and return the constant varieties admitting at least one solution in the dependents.

The organizing intuition — **independents universally quantified, dependents and constants existentially quantified** — is semantically correct, and §4 will keep it. But the operational recipe “collect over the independents first, then eliminate the dependents inside the coefficients” fails on both sides, and the failures are not edge cases:

Example 1 (over-constraint: the defining equation). *Apply the recipe to the ansatz equation $v - v_1x - v_2y - v_3z = 0$. Grouping by x, y, z gives four coefficient rows: v (the constant term), $-v_1$, $-v_2$, $-v_3$. Three rows contain no dependent variable at all, so the existential step has nothing to witness and the recipe forces $v_1 = v_2 = v_3 = 0$ — destroying a perfectly good defining equation of the ansatz. (The v -row is “witnessed” by $v = 0$, compounding the damage.)*

Example 2 (under-constraint: the frozen witness). *Apply the recipe to the combined toy system $A : \Psi'' - c\Psi$, $P : \Psi' - \Psi$ without the Phase-I closure (or, equivalently, apply it directly to Algorithm 1’s reduced remainder $r = \Psi' - \Psi$). No independent variable appears, so there is a single coefficient row $\Psi' - \Psi$, containing dependents only; the existential step asks whether some values (ψ_0, ψ_1) satisfy $\psi_1 - \psi_0 = 0$ — always yes. The recipe returns all of $\mathbb{C}$, but $\mathcal{E} = \{c = 1\}$ and $\mathcal{M} = \emptyset$. The recipe is not computing the existence locus: it has quantified over frozen values of Ψ, Ψ' , constant in x , whereas a solution’s jets vary with x — the constraint that kills $c \neq 1$ is the derivative of the imposed relation, $D(\Psi' - \Psi) = \Psi'' - \Psi' \rightarrow c\Psi - \Psi = (c - 1)\Psi$, which only differential closure supplies.*

Example 1 over-constrains ($\{v_1 = v_2 = v_3 = 0\}$ where the true answer is “no condition”); Example 2 under-constrains ($\mathbb{C}$ where the true answer is $\{1\}$). So the naive Phase III computes a locus that is in general *incomparable* to both $\mathcal{M}$ and $\mathcal{E}$: it is the set of constants for which some x -constant assignment of the dependent symbols annihilates the equations identically in the independents — a hybrid with no differential meaning. The next two sections diagnose why, and what the correct discipline is.

4 The quantifier principle, made precise

4.1 The four classes of symbols

Fix a leaf system of Phase II: a differentially triangular set T with inequations NE , under a ranking $<$ on the jets. The non-constant symbols fall into four classes, and *each class admits exactly one sound projection operation*:

Definition 1 (symbol classes). *Relative to $(T, <)$:*

1. **Constants** c : the answer space. Never quantified away; the output lives in $K[c]$.
2. **Independent variables** x, y, z (and their algebraic extensions, here r with $r^2 = x^2 + y^2 + z^2$): universally quantified — an equation must hold at every point of some open set.

3. **Principal jets:** the leaders of T and all their derivatives. Each is determined by its equation (solved form; the initial is a unit by NE): its quantifier is irrelevant because it has a unique witness.
4. **Parametric jets:** the remaining jets — the free Taylor data of the family cut out by T (finite in number for a finite-type system). This class is the selector: quantify it universally and the projection computes membership; existentially, existence.

For the hydrogen ansatz under the paper’s ranking $\Psi'' > \Psi' > \Psi > v > r > \text{constants} > x, y, z$: the principal jets are Ψ'' , v , r (leaders), the spatial derivatives Ψ_x , Ψ_y , Ψ_z (leaders of the chain rules), and all their derivatives; the parametric jets are exactly Ψ, Ψ' . It is no accident that the paper’s projection collects like terms in precisely x, y, z, r, Ψ, Ψ' — classes 2 and 4 — and in nothing else.

4.2 The two implementations of $\forall$, and the one of $\exists$

Principle 1 (operations, not just quantifiers). *Universal quantification is implemented by coefficient collection; existential quantification by elimination; determined symbols by substitution (triangular rewriting). Collection is sound only over symbols algebraically independent of one another and of everything retained; elimination over the jets is sound only after differential closure.*

We justify the three clauses in turn.

Collection ($\forall$). An equation $e \in K[c][x, y, z, r][\text{parametric jets}]$, containing no principal jet, vanishes for all real values of the independents and all admissible values of the parametric data iff every collected coefficient $p_\alpha(c)$ in (4) vanishes. Two ingredients:

- *Syntactic:* distinct power products are linearly independent over $K[c]$, so $e \equiv 0$ as a polynomial iff all $p_\alpha = 0$. For the algebraic extension r this requires e to be reduced modulo $r^2 - (x^2 + y^2 + z^2)$ first (r -degree ≤ 1); the reduced monomials are linearly independent as functions on the quadric. This is one of several places where “fully reduced” is load-bearing.
- *Semantic (density):* vanishing for all real values is equivalent to identical vanishing because $\mathbb{R}^3$ is Zariski-dense in $\mathbb{C}^3$; vanishing for all admissible parametric data is equivalent because, on a cell with its inequations, the parametric Taylor data is freely assignable at every admissible base point (Riquier existence — for the hydrogen class, elementary second-order-ODE existence), so the realized values of $(x, y, z, r, \Psi, \Psi')$ fill a Zariski-dense subset of the quadric-times-jet space.

Note what collection is *not*: it is not elimination. One does not eliminate a universally quantified variable from an ideal — elimination computes the projection of a variety, which is the $\exists$ semantics. Collection is the linear-algebra operation dual to it, and this asymmetry is why the constants can stay in the coefficient field while the universal variables stay symbolic.

Elimination ($\exists$). For the existence reading, the object is the prolonged, algebraized, un-saturated ideal of the leaf system, $J = (T^{[N]}) \subseteq K[\text{jets}_{\leq N}, x, y, z, r, c]$, at closure order N . Lange-Hegemann’s Theorem 4.1 [13] (equivalently Lemma 3 of [3]) says that for an involutive system, every point of $V(J)$ respecting the inequations is the truncation of a genuine formal solution. Therefore

$$J_c = \left(J : (\Psi \cdot \prod_{h \in NE} h)^\infty \right) \cap K[c] \quad (6)$$

cuts out the Zariski closure of the cell’s contribution to $\mathcal{E}$: eliminate *all* the jets *and the base point* x, y, z, r (a solution needs only *some* point of analyticity — demanding all points would wrongly exclude, e.g., the Coulomb singularity $r = 0$), after quotienting out the zero section and localizing at the cell’s inequations. The universal “for all points near the base point” is carried entirely by the prolongation generators $\theta a \in T^{[N]}$: prolongation is the Taylor-basis implementation of $\forall$ -over-independents, dual to the monomial-basis implementation by collection. Example 2 is exactly what happens when this clause is ignored: without the generators $D^k(\Psi' - \Psi)$, the eliminated ideal knows nothing of $(c - 1)\Psi$ and the projection overshoots.

Substitution (determined). A principal jet is rewritten by its solved-form equation — this is what Ritt reduction *is*. Its equation is a definition, not a constraint row; its unique witness is supplied by the triangular structure. Feeding a defining equation into either projection operation is the category error of Example 1.

4.3 The two sound pipelines

Assembling Principle 1, there are exactly two sound orders of operations — one per quantifier assignment on the parametric jets — and the naive Phase III is sound in neither reading because it interleaves them [6, Finding 3: “exactly one $\forall$ algorithm and one $\exists$ algorithm”]:

	membership $\mathcal{M}$ (per cell)	existence $\mathcal{E}$ (per cell)
1. differential closure	full reduction of P by T	prolongation of $T \cup \{P\}$ to passivity
2. principal jets	substituted away by the reduction	eliminated (they are determined)
3. parametric jets	collected (universal)	eliminated (existential, after 1!)
4. independents x, y, z, r	collected (universal)	eliminated ($\exists$ base point; $\forall$ carried by 1)
5. constants	land in $K[c]$: the system $\{p_\alpha(c)\}$	land in $K[c]$: the ideal J_c of (6)

In the membership column no coefficient containing a dependent symbol is ever “set to zero”: collection continues *through* the parametric jets until the coefficients are pure constants. In the existence column nothing is ever collected: every non-constant is eliminated, and the universal content lives in the prolongation generators. The trap is precisely the hybrid — collect over the independents, stop, and existentially witness the jet-carrying coefficients — because collection over x silently reclassifies every dependent symbol as a constant-in- x coefficient, which dependents are not (they are functions of x ; that is what “dependent” means).

Remark 1 (relation to comprehensive Gröbner systems and QE). *The $\forall\exists$ block structure over an algebraically closed field is a quantifier-elimination problem, and the tools align: the $\exists$ block is constructible-image computation (elimination ideals, per parameter cell — the comprehensive Gröbner system / Gröbner cover technology [21, 22]), while the $\forall$ block, because it is universal over a polynomial identity, degenerates to coefficient collection — linear, requiring no cell analysis and no real QE. This is why the whole pipeline stays inside Gröbner-land. The only real-vs-complex subtlety sits in the $\exists$ block: elimination certifies a complex witness; realness of solutions is a separate question the paper handles at the level of the final varieties.*

5 Resolution of the $v = v_1x + v_2y + v_3z$ puzzle

5.1 The distinguishing criterion

Criterion 1. *An equation may participate in independent-variable coefficient-zeroing if and only if it contains no principal jet — equivalently, iff it is fully reduced with respect to the cell’s triangular set — and the collection must then run over the independents and the parametric jets together, terminating in coefficients built from constants alone. An ansatz-defining equation never qualifies: its leader is a principal jet by definition — the equation is the solved form that defines that jet — so it enters the projection only as a rewrite rule (membership reading) or as an elimination generator (existence reading), never as a constraint row to be collected.*

Applied to the puzzle: in $v - v_1x - v_2y - v_3z = 0$ the symbol v is a dependent variable — itself a function of x, y, z — and under any admissible ranking it is the equation’s leader (it is the only jet present). The equation is therefore a definition of v , and Criterion 1 excludes it from collection in both readings:

- *Membership reading (Algorithm 1):* v is principal, so full reduction substitutes $v \mapsto v_1x + v_2y + v_3z (+ v_4r)$ everywhere in the remainder before the projection ever looks at it. The defining equation is consumed as a rewrite rule; v never appears among the collection variables — witness the paper’s collection set x, y, z, r, Ψ, Ψ' , from which v is conspicuously absent while the *parametric* jets Ψ, Ψ' are present.
- *Existence reading (bounded prolongation):* the equation and its prolongations $v_x - v_1, v_y - v_2, v_z - v_3, v_{xx}, \dots = 0$ enter the ideal J , and the v -jets are *eliminated*. The elimination ideal $(v - v_1x - v_2y - v_3z, v_x - v_1, \dots) \cap K[v_1, v_2, v_3]$ is zero: no condition on the constants, as it must be.

Either endpoint is safe. The naive Phase III fails because it stops the universal collection at the independents, leaving “coefficients” that still contain dependent symbols, and then treats those coefficients as objects that must vanish (rows without dependents — forcing $v_1 = v_2 = v_3 = 0$) or be witnessed by frozen values (rows with dependents — Example 2). Both failure modes are the same error seen from two sides: *collection over x, y, z presupposes that everything left in a coefficient is constant in x, y, z , and dependent variables are not.*

5.2 What distinguishes the defining equation from the reduced remainder

Both $v - v_1x - v_2y - v_3z$ and a reduced remainder r are elements of the same differential ring; both may contain jets; the remainder $r = \Psi' - \Psi$ of the toy example even has a jet *leader* (Ψ'). The distinction is not “contains a jet” but the *status of its jets relative to the triangular set doing the projecting*:

- the defining equation contains its own leader — a jet that is principal *for the ansatz* — so relative to A it is a definition;
- the fully reduced remainder contains, by the definition of full reduction, no leader of A and no derivative of one: every jet in it is *parametric* for A , i.e. free family data over which the membership question universally quantifies.

The ranking is what decides which jet of each equation is its leader, hence which jets are principal, hence which symbols the collection may treat as free. Change the ranking and the same polynomial

can change roles — which is not a defect but the design: the ranking of the paper (§Ranking of [1]) is engineered exactly so that Ψ'' , the spatial derivatives, v , and r are the defined symbols and Ψ, Ψ' the free ones.

The “equal footing” aspiration is then realized without contradiction: adjoining P to A and decomposing means the remainder itself *becomes a defining equation* of the combined system — in the toy example, $\Psi' - \Psi$ acquires the leader Ψ' , turning Ψ' from parametric (for A) to principal (for $A \cup \{P\}$), and the completion then derives $(c - 1)\Psi$, whose *initial* $(c - 1)$ carries the constant condition as a Thomas split, not as a coefficient-zeroing. On an equal footing, constants conditions materialize as *initials of derived equations* and as *elimination ideals of cells* — never by zeroing jet-carrying coefficients. Coefficient-zeroing reappears only when, a cell having been fixed, one asks the membership question *of that cell* — at which point its defining equations act again as rewrite rules and Criterion 1 applies verbatim.

Remark 2 (why the physics tradition never met the puzzle). *The ansatz of [23] (and of the quasi-exact-solvability literature it implements) is explicit: $\psi = g(r)e^{y(r)}$ is substituted into the wave equation before any algebra happens, so every dependent symbol is eliminated by construction and raw power-equating is Criterion 1-compliant without anyone having to say so. The puzzle is a phenomenon of implicit (differentially triangular) ansätze, where definitions coexist with constraints in one system and must be told apart — by their leaders.*

5.3 The criterion tested on the partial-strata examples

Example A ($A : \Psi'' - c\Psi$, $P : \Psi' - \Psi$; **one derivation**). *Membership pipeline:* P is already fully reduced w.r.t. A (its leader Ψ' ranks below Ψ'' , and no derivative of Ψ'' appears); its jets Ψ, Ψ' are parametric for A ; collect over them: coefficients $+1, -1$; $\mathcal{M} = V(1, -1) = \emptyset$. ✓ (matches [2, Ex. 1]). *Existence pipeline:* close: $(A \cup \{P\})^{[2]} = \{\Psi'' - c\Psi, \Psi' - \Psi, \Psi'' - \Psi'\}$; auto-reduce: $\Psi'' - \Psi' \rightarrow c\Psi - \Psi' \rightarrow (c - 1)\Psi$, tracking no division (all initials are 1 until the derived equation appears); the derived equation’s initial $(c - 1)$ splits Phase-II-style: on $c \neq 1$ the cell forces $\Psi = 0$ (trivial; discarded by the $:\Psi^\infty$ quotient), on $c = 1$ the cell is $\{\Psi' = \Psi\}$ with parametric jet Ψ ; eliminating jets with $\Psi \neq 0$ leaves the ideal $(c - 1)$, i.e. $\mathcal{E} = \{1\}$. ✓ The dimension criterion agrees: the $c = 1$ cell has one parametric jet where A has two — a proper subfamily, hence a partial stratum, hence in $\mathcal{E} \setminus \mathcal{M}$.

Example B ($A : \Psi_{yy} - c\Psi$, $P : \Psi_x - \Psi$; **two derivations**). Here $A \cup \{P\}$ is already autoreduced and coherent: the Δ -polynomial of the leader pair (Ψ_x, Ψ_{yy}) reduces to zero, so Phase I terminates with no derived equation and Phase II never splits — one cell, equations $\{\Psi_x - \Psi, \Psi_{yy} - c\Psi\}$, parametric jets $\{\Psi, \Psi_y\}$. *Membership:* reduce P by A : remainder P itself; collect over the parametric data of A (which is infinite — A alone is not finite type; the collection over Ψ_x, Ψ suffices): coefficients $+1, -1$; $\mathcal{M} = \emptyset$. ✓ *Existence:* eliminate the jets from the closed cell with $\Psi \neq 0$: the prolongations $(\Psi_{xy} = \Psi_y, \Psi_{xyy} = \Psi_{yy} = c\Psi$ from both routes, ...) are consistent for every c ; the elimination ideal in $K[c]$ is (0) ; $\mathcal{E} = \mathbb{C}$ — every c is a partial stratum, the stratum being the subfamily $e^x(K_1 e^{\omega y} + K_2 e^{-\omega y})$. ✓ (matches [2, Ex. 2]). The dimension criterion again separates the readings: the cell’s parametric set $\{\Psi, \Psi_y\}$ is a collapse from A ’s infinite parametric set (two arbitrary functions $F(x), G(x)$ down to two constants), so no cell is dimension-preserving and $\mathcal{M} = \emptyset$ — even though the decomposition never split and the adjoined equation “reduced to nothing new,” confirming that membership cannot be read off the splitting combinatorics, only off the quantifier discipline (or dimensions).

The puzzle equation itself. Worked in §5: no condition on v_1, v_2, v_3 in either pipeline; $v_1 = v_2 = v_3 = 0$ arises only from the unsound hybrid. ✓

6 The corrected algorithm, and which locus it computes

6.1 Statement

Algorithm (prolongation–projection, corrected). *Input:* ansatz $A(c)$, PDE P , ranking $<$. *Output:* the existence locus $\mathcal{E}$ stratified into cells, with each cell’s membership status decided; hence both loci.

- I. Close and track.** Run the completion loop on $A \cup \{P\}$ (prolong $\rightarrow$ auto-reduce $\rightarrow$ adjoin nonzero residues) to passivity, tracking every initial/separant used as a divisor (5). No a-priori order bound is assumed (prolong-until-stable; the ansatz-only bound $N_0(A)$ is *not* valid for the combined system).
- II. Split and recurse.** For each tracked divisor h_i : keep $h_i \neq 0$ on the current branch; spawn a branch with $h_i = 0$ adjoined and recurse from I. Collect the leaves (T_j, NE_j) — finitely many, each passive.
- III_∃. Project each leaf onto the constants (existence).** Algebraize (jet $\mapsto$ indeterminate) at the leaf’s closure order; compute $J_c^{(j)} = (J_j : (\Psi \prod_{h \in NE_j} h)^\infty) \cap K[c]$, eliminating all jets and the base point. Then $\bar{\mathcal{E}} = \bigcup_j V(J_c^{(j)})$, each leaf contributing its stratum (bad-locus strata included — nothing was saturated by H_A globally).
- III_∀. Decide membership per leaf (optional selector).** On each leaf, either (a) fully reduce P by T_j and collect over independents *and* the leaf’s parametric jets, testing whether the leaf’s constant relations already imply every collected coefficient (radical membership, as implemented in `joca-rg-combined.sage`); or equivalently (b) test dimension preservation: the leaf’s parametric-derivative set equals the ansatz’s. Leaves passing are membership cells; their union recovers $\mathcal{M} = \mathcal{V}$; the rest are partial strata.

The changes from the proposal of §3 are exactly two, both forced by §4: Phase I is a completion loop rather than a one-shot prolongation to $N_0(A)$; and Phase III’s two quantifier moves are *unmixed* — elimination of everything (III_∃) or collection through everything non-constant (III_∀), never collection-then-elimination on the same equation.

6.2 Which locus — and the selector

Phase III_∃ computes, per leaf, the Zariski closure of that leaf’s contribution to the existence locus; the union over leaves is $\bar{\mathcal{E}} = \overline{\mathcal{V} \cup \{\text{partial strata}\}}$ [3, §2bis(b), superset remark]. It is a *new route to the joint-variety projection*, not to a third locus: by the two-pipelines analysis there is no third quantifier assignment to compute. What is genuinely selectable is the per-cell refinement:

- **Existence per cell (III_∃):** keeps the bad-locus strata (hydrogen $E = -1/2$, $E = -1/8$ cells) that Algorithm 1’s $h \neq 0$ hypothesis fences off — the whole point of the equal-footing treatment.

- **Membership per cell** ($\text{III}_\forall$): the *dimension-preservation criterion* of [2, §The correct selection criterion] — a cell belongs to $\mathcal{V}$ iff the PDE removed nothing from the family, i.e. iff the cell’s parametric set is the ansatz’s. Lange-Hegermann’s differential counting polynomial [13] makes this effective by inspection of the cell (count the parametric derivatives); the radical-membership test of `joca-rg-combined.sage` is the implemented equivalent. Note the test must be the *radical-membership* form, not “re-reduce P by the cell’s chain and collect”: reduction against a chain’s bare equations does not saturate, and the hydrogen `joca-rg.sage` run shows the resulting projection re-admitting the spurious prime (a_0, a_1) while absorbing the genuine strata inside it [10].

So the corrected algorithm is a *selectable hybrid*: $\text{III}_\exists$ alone gives $\mathcal{E}$; $\text{III}_\exists$ filtered by $\text{III}_\forall$ gives $\mathcal{V}$; reporting both, with each partial stratum labeled by its cell, gives strictly more information than either endpoint algorithm — which is the honest way to state its value over Algorithm 1 (which cannot see the strata) and over a raw combined decomposition (which cannot tell the strata from $\mathcal{V}$ without exactly this selector).

6.3 Relation to the existing methods

Phases I–II are a differential Thomas decomposition with the split set generated lazily by the reduction trace — a strategy, not a new decomposition notion; the guarantees (disjointness, passivity of leaves) are Thomas’s [11], and any correctness claim should lean on that literature rather than re-prove it. Phase $\text{III}_\exists$ per leaf is the bounded-prolongation route of [3] run per cell (where its two known pathologies are mildest: on a cell the chain is triangular, the inequations are explicit, and the closure order is the cell’s own). Phase $\text{III}_\forall$ is Algorithm 1 run relative to a cell, with the saturation handled by radical membership. The prolong-to-order- i / algebraize / eliminate construction is classical (it is inside [17], there with the H -saturation we deliberately omit); the parametric splitting is comprehensive-Gröbner technology [21, 18]. What this note adds is not machinery but the *quantifier discipline* — the four-class table of Definition 1, the two-pipelines theorem-in-outline of §4, and Criterion 1 resolving the defining-equation puzzle — which is exactly the piece a correct implementation of Brent’s Phase III sketch was missing.

7 The staged route: ansatz first, PDE per cell

The algorithm of §6 runs the completion loop on $A \cup \{P\}$. In practice that is exactly what does not terminate. This section develops the alternative that *does* run — decompose the ansatz alone, then bring in the PDE cell by cell — says precisely what it loses, and shows that the lost half can be recovered by a *fixed-chain prolongation* that never re-enters the differential engine.

7.1 The staged route, and what it keeps

Staged route. Run a differential Thomas decomposition of the *ansatz alone*: $\text{Sol}(A) = \bigsqcup_j \text{Sol}(T_j, NE_j)$, each cell passive. Then, per cell: fully reduce P modulo T_j , obtaining $h_j P \equiv \rho_j \pmod{[T_j]}$, and project ρ_j onto the constants.

This is not hypothetical: it is `joca-thomas.sage`, whose output `joca-thomas-openmaple.out` records all 29 hydrogen cells with their projections and minimal primes [7]. For the universal reading it loses nothing, and it gains the bad locus:

- *Completeness of the partition.* Thomas's cells are disjoint and exhaustive, so $\text{Sol}(A) \cap \text{Sol}(P) = \bigsqcup_j (\text{Sol}(T_j) \cap \text{Sol}(P))$ exactly. Decomposing A first discards nothing about A .
- *The remainder is the whole story (for $\forall$).* Each T_j is *passive*, which is precisely the hypothesis Rosenfeld's lemma needs: Ritt reduction to zero decides membership in $[T_j] : H_{T_j}^\infty$. No prolongation of P is required for the membership question.
- *Criterion 1 is satisfied for free.* ρ_j is fully reduced modulo T_j , hence contains no principal jet of the cell, hence qualifies for collection; the cell's defining equations enter only as rewrite rules, never as constraint rows. The v -puzzle cannot arise.
- *The $h \neq 0$ hypothesis holds cell-by-cell.* The Ritt denominator h_j is a product of initials and separants of T_j 's equations — and in a Thomas *simple system* those are exactly the cell's inequations NE_j . So $h_j \neq 0$ on the cell by construction. Theorem 1's completeness hypothesis, which Algorithm 1 can assert only on the generic branch, is satisfied on *every* cell, and Algorithm 2's bad-locus recursion is subsumed by the ansatz decomposition. This is why cell 26 (`zero {a0,a1}`) exists at all, and why it hands back the 2s stratum at $E = -1/8$ that `joca.sage`'s five primes miss.

One terminating Thomas run on A (~ 18 min) thus buys the completeness that the non-terminating combined run was supposed to buy.

7.2 What it loses: exactly the parametric-leader strata

Proposition 1 (the staged route's blind spot, exactly located). *Let (T_j, NE_j) be a cell of the Thomas decomposition of A , and ρ_j the full Ritt remainder of P modulo T_j . Exactly one of three cases holds, and only the third can hide a stratum:*

1. $\rho_j = 0$: *the cell is a membership cell throughout; $\mathcal{E}_j = \mathcal{M}_j$. Nothing is lost.*
2. $\rho_j \neq 0$ but contains no parametric jet of T_j (only constants and x, y, z, r): *collection over the independents lands directly on constant equations, and there is no witness freedom for $\exists$ to exploit; $\mathcal{E}_j = \mathcal{M}_j$. Nothing is lost.*
3. $\rho_j \neq 0$ and contains at least one parametric jet of T_j : *this is where — and only where — $\mathcal{E}_j \setminus \mathcal{M}_j$ can be nonempty, and where projecting the bare remainder universally discards information.*

Justification. ρ_j is fully reduced modulo T_j , so by the definition of full reduction it contains no principal jet of the cell: every jet occurring in it is parametric for T_j . Hence if $\rho_j \neq 0$ and carries a jet at all, then in the combined system $T_j \cup \{P\}$ the equation $\rho_j = 0$ has as its leader a *formerly parametric* jet, promoting it to principal. That promotion *is* the stratum: cutting a proper nontrivial subfamily out of a cell means imposing an algebraic condition on the cell's free Taylor data, and a new equation with a parametric leader is the only mechanism by which that can happen. Conversely, in cases 1 and 2 no jet is promoted, the free data is untouched, and every solution of the cell either solves P or the cell's constants are constrained outright. $\square$

The consequence deserves stating flatly, because it is easy to be misled by hydrogen's good luck:

Principle 2 (which strata the staged route sees for free). *Decomposing A splits on degeneracies of A 's own description. The combined decomposition would additionally split on degeneracies produced by P 's interaction with A . The staged route sees the first kind as cells and can never see the second kind at all.*

The $E = -1/8$ stratum is of the *first* kind: it lives on **zero** $\{\mathbf{a0}, \mathbf{a1}\}$, a degeneracy of the ansatz’s own leading coefficient, which A ’s decomposition carves out unaided — and on that cell the universal projection legitimately returns it. There is no reason to expect this in general, and no reason to think it exhausts hydrogen’s strata. A stratum whose split arises only *after* P is adjoined — an initial like $(c - 1)$ in Example 2, which exists nowhere in A — has no cell to live in, and the universal projection on the cell containing it returns nothing.

7.3 Why re-completing $T_j \cup \{P\}$ per cell hits a wall

The obvious repair is to adjoin P to each cell and re-run the differential Thomas decomposition there. This has been implemented and measured (**DifferentialThomas-sage**), and it does not work on 5 of the 29 hydrogen cells — namely cells 1, 2, 25, 26, 28 [8]. The diagnosis is worth recording, because it is what motivates §7.4:

- *Not coefficient swell.* Over 232 sampled pseudo-division steps in cell 1 the maximum coefficient bit-length was **16 bits**; no step exceeded 64 bits. Ruled out by measurement.
- *Not (any longer) retained garbage.* BLAD’s stack allocator never rolled back, so every intermediate of every pseudo-division stayed allocated for the run’s lifetime. An in-epoch arena GC fixed this [9]: hydrogen peak RSS fell from 6910 MB to ~ 1.75 GB.
- *The residual wall is intra-step live-set swell inside a single **pseudo_remainder** call.* After the GC fix the failures changed character rather than disappearing: cells 1/2/26/28 now *time out at 6 h while compute-bound* with flat RSS, and cell 25 dies with a clean **MemoryError** on a genuine live set at 38.7 GB. Branch tracing records **zero** staging steps across those six hours — the engine never gets out of one reduction. The peak remainder in cell 1 carried **218,488** monomials.

The decisive observation is that *those same five cells are cells on which the staged projection completes without difficulty*: cell 1 (generic), cell 2 (**zero** $\{\mathbf{v4}\}$), cell 25, cell 26, cell 28 all appear in **joca-thomas-openmaple.out** with their remainders reduced and their primes computed. So computing $\rho_j = P \bmod T_j$ is *cheap*. What explodes is not the reduction of P by T_j ; it is the *re-completion* of $T_j \cup \{P\}$.

The reason is structural, and Proposition 1 already named it. Adjoining ρ_j promotes a parametric jet to a leader. The chain is therefore no longer passive; the engine must recompute all Δ -polynomials against the new leader, split on the initials, separants and discriminants of every derived equation, branch, and re-reduce inside every branch — all in a jet space that *grows* with each prolongation. Every one of those is a cost the staged route was supposed to have already paid.

7.4 The fixed-chain prolongation

The repair is to stop treating ρ_j as an equation to be added to a chain, and treat it as an *element to be prolonged against a chain that never changes*. The cell is left exactly as Thomas produced it — passive, triangular, untouched — and used purely as a rewriting system.

Lemma 1 (a cell is a finitely generated differential algebra). *Let (T_j, NE_j) be a passive cell with parametric jets U_j (finite, for a finite-type system), and set*

$$R_j = K[c][x, y, z, r]/(r^2 - x^2 - y^2 - z^2) [U_j].$$

Then $\partial_x, \partial_y, \partial_z$ extend to commuting derivations of the localization $R_j[NE_j^{-1}]$, computable by Ritt reduction against T_j .

Justification. For $u \in U_j$, the jet $\partial_x u$ is either principal for T_j — in which case the chain rewrites it, in solved form, as an element of R_j , legally, because the initial it divides by lies in NE_j — or parametric, in which case it lies in U_j already. Passivity is what makes the rewriting confluent, hence the extension well defined; the inequations are what make the divisions legal. $\square$

For the generic hydrogen cell $U_j = \{\Psi, \Psi'\}$, so R_j has just *six* non-constant generators (v is principal, substituted away by $v = v_1x + v_2y + v_3z + v_4r$), and the derivations are explicit:

$$\begin{aligned} \partial_x x &= 1, & \partial_x r &= x/r, & \partial_x v &= v_x = v_1 + v_4 x/r, & \partial_x \Psi &= \Psi' v_x, \\ \partial_x \Psi' &= \Psi'' v_x = -\frac{(b_0 + b_1 v) \Psi' + (c_0 + c_1 v) \Psi}{a_0 + a_1 v} v_x. \end{aligned} \quad (7)$$

Every denominator that appears — $(a_0 + a_1 v)$, the ODE's initial, and r , the separant of $r^2 - (x^2 + y^2 + z^2)$ — is an inequation of the cell. That is what passivity buys, and it is the whole content of Lemma 1.

Now build the ascending chain of *commutative* ideals in R_j :

$$I^{(0)} = (\rho_j), \quad I^{(k+1)} = \left(I^{(k)} + \partial_x I^{(k)} + \partial_y I^{(k)} + \partial_z I^{(k)} \right) : (\Psi \prod_{h \in NE_j} h)^\infty, \quad (8)$$

and on stabilization return $I^{(\infty)} \cap K[c]$: the cell's contribution to $\bar{\mathcal{E}}$. Termination is the Hilbert basis theorem — the chain ascends in a Noetherian ring.

The saturation must be taken *before* the closure test, not after: what must be closed under the derivations is the saturated ideal. This is sound — Kolchin's lemma says a radical differential ideal saturated by any multiplicative set is again a radical differential ideal — but it dictates the implementation: saturate by ideal quotient, **not** by a Rabinowitsch variable $t\Psi - 1$. A Rabinowitsch t is not a jet, the derivations of Lemma 1 do not extend to it, and the very structure that makes the method work would be destroyed.

7.5 Stabilization without Δ -polynomials

Criterion 2 (D -closure). *Let G be a Gröbner basis of $I^{(k)}$ in R_j . Stop iterating (8) when*

$$\partial_x g, \partial_y g, \partial_z g \xrightarrow{G} 0 \quad \text{for every } g \in G.$$

Then every higher derivative $D^k \rho_j$ already lies in $I^{(k)}$, so the elimination ideal $I^{(k)} \cap K[c]$ is final.

This is a *certificate*, not a heuristic, and the distinction matters. Watching the elimination ideal $I^{(k)} \cap K[c]$ ascend and appear to stop moving is a useful cheap progress monitor, but it proves nothing: the elimination ideal can sit still for one prolongation and then jump. D -closure of the full ideal proves closure. Use the first to decide whether to keep going, the second to decide that you are done.

Criterion 2 carries the same mathematical content as a Δ -polynomial test — it must; integrability is integrability — but it is a different *computation*, and the three differences are precisely the three things that blow up in §7.3:

1. **The cell is never tested against itself.** T_j is *already passive*: Thomas established that when it decomposed the ansatz, and all Δ -polynomials among the ansatz equations already reduce to zero. They are never recomputed, because T_j is never modified. Handing $T_j \cup \{P\}$ back to Thomas, by contrast, changes the leader structure and forces *every* pair to be reprocessed.

2. **There are no pairs.** A Δ -polynomial $S_q \theta_p p - S_p \theta_q q$ is a pairwise, leader-matched combination, and it exists precisely to avoid dividing by leading coefficients that might vanish. On a passive cell the leading coefficients *are units* — they are the inequations — so the division is legal and the entire leader-matching apparatus collapses to plain differentiation: apply ∂ to a generator, reduce, done. One generator in, one polynomial out. No pair queue, no leader analysis.
3. **Nothing acquires a leader, so nothing can split.** Thomas splits whenever the initial or separant of a *derived* equation may vanish. Here the $D^k \rho_j$ are never made into chain equations; they are ideal generators, they have no initials, and there is nothing to split on. The vanishing conditions Thomas would have branched on reappear instead as *components* of $V(I^{(\infty)} \cap K[c])$ — which is where they were wanted all along: as answers, not as branches.

And the variable set is fixed. In the differential engine the jet space grows with every prolongation; here ∂g lands back in the same six-generator ring, always, by Lemma 1. The generators may grow in *degree* — each pseudo-division multiplies by $(a_0 + a_1 v)$ and by r , so degrees grow roughly linearly in the prolongation order — but never in *variable count*. That is a structural reason to expect the 218,488-monomial swell not to recur, and it is the reason to expect the fixed-chain route to reach the five cells that defeat re-completion.

7.6 The linear-homogeneous structure: existence is a rank condition

For the hydrogen class — and for the whole linear-homogeneous class the paper treats — there is a further structural fact that collapses the $\exists$ computation, and it was found by running the algorithm rather than by thinking about it. The ansatz and the PDE are *linear homogeneous* in Ψ . Reduction and the derivations of Lemma 1 are linear operations, so:

Proposition 2 (the generators are linear forms in the parametric jets). *Let $u_1, \dots, u_m$ be the parametric jets of the cell. Then ρ_j and every $\partial^k \rho_j$ is a linear form*

$$\rho^{(i)} = \sum_{l=1}^m M_{il}(c, x, y, z, r) u_l,$$

with the coefficients M_{il} free of every jet. Consequently, writing M for the coefficient matrix,

$$\mathcal{M}_j = \{c : M \equiv 0 \text{ identically}\} \quad (\text{every entry vanishes — the paper's coefficient collection}) \quad (9)$$

$$\mathcal{E}_j = \{c : \text{rank } M(c, p) \leq m - 1 \text{ for some base point } p\} \quad (\text{a nontrivial kernel } (u_1, \dots, u_m) \neq 0 \text{ exists}). \quad (10)$$

Three things follow at once. First, $\mathcal{M}_j \subseteq \mathcal{E}_j$ is now trivial: $M \equiv 0$ makes every minor vanish everywhere. Second, the $m \times m$ minors of M *contain no jets at all*, so the jet variables leave the elimination for free and only the base point x, y, z, r remains to be eliminated. Third — and this is the part that matters operationally — the two loci are distinguished by *how* the minors vanish: identically (membership) versus on an isolated base-point locus (a partial stratum). The quantifier selector of §6 is, for this class, a rank test.

Remark 3 (what the parametric set actually is, per cell). *The parametric jets are read off the cell, not assumed. On the generic cell they are $\{\Psi, \Psi'\}$, as the paper's collection set implies. On*

cells 26/27 ($a_0 = a_1 = 0$) the ODE loses its Ψ'' term altogether, so Ψ' becomes the leader and Ψ'' goes free: the parametric set is $\{\Psi, \Psi''\}$. Still $m = 2$, still 2×2 minors — but an implementation that hard-coded $\{\Psi, \Psi'\}$ from the generic cell would silently collect over a determined symbol and over-constrain, which is Criterion 1 violated at one remove.

7.7 Passivity is a commutator condition — and must be checked

Everything in §§7.4–7.6 rests on Lemma 1, and the lemma rests on *passivity*. It is worth knowing that this hypothesis is cheaply *testable* inside the same machinery, because in practice it is the thing that goes wrong.

Criterion 3 (the derivations must close on the ring). *Build the derivation table: for every parametric jet $u \in U_j$ and every derivation ∂_d , reduce $\partial_d u$ modulo the chain. Lemma 1 says that on a passive cell the result is always an element of R_j — the derivative of a parametric jet is either principal (rewritten by the chain) or parametric (already in U_j). If the reduction instead throws up a jet that is not a generator of R_j , the construction is invalid: the chain is not passive, the parametric set is really infinite, and the fixed-chain route must refuse. The check costs one reduction of a single symbol per (u, d) pair — milliseconds — and it is the guard the implementation must carry.*

If it is skipped, the route does not fail loudly; it *lies*. The “parametric” jets are not free, the coefficient matrix M of Proposition 2 is built over the wrong column set, and the rank test returns a confident wrong answer.

Remark 4 (a live defect in `joca-thomas.sage`, and how it was caught). *The script builds each cell’s chain by substituting the cell’s vanishing constants into the raw ansatz. That is the completed Thomas chain only on cells where the ansatz was already passive — the generic cell. On the degenerate strata the real Thomas cells carry extra integrability equations, and the re-specialized ansatz does not. Concretely on cells 26/27 ($a_0 = a_1 = 0$) the ODE loses its Ψ'' term altogether, so Ψ'' is unconstrained — and so are $\Psi''_x, \Psi''_{xx}, \dots$. The parametric set is infinite, the cell is not of finite type with respect to this chain, and Criterion 3 catches it precisely:*

$$\text{reduce}(\partial_x \Psi'') \longrightarrow \dots \Psi''_x \dots \quad \text{— a jet that is not a generator of } R_j.$$

Two false trails are worth recording, because both looked like results. First, run without the guard, the route cheerfully reports the parametric set of cells 26/27 as $\{\Psi, \Psi''\}$ and returns an elimination ideal of (0) . Second, the tempting “passivity = commuting derivations” test ($[\partial_i, \partial_j](u) = 0$, the Δ -polynomial condition transplanted) is vacuous here: it reports these cells *PASSIVE*, because the out-of-ring jet is silently mangled before the commutator is ever formed. Only the ring-closure check of Criterion 3 is decisive.

All of this is harmless for the universal route — Ritt reduction still reduces the PDE fully, which is why the recorded 29-cell projection and its $E = -1/8$ prime stand — and fatal for the existential one. It is the general rule in the sharpest possible form: $\forall$ **needs only reduction**; $\exists$ **needs closure**.

Remark 5 ($\text{rank} \leq m - 1$ is not membership). *A related trap, met the same way. “All $m \times m$ minors of M vanish identically” means $\text{rank } M \leq m - 1$ everywhere, hence a 1-dimensional kernel at every c : a proper subfamily solves for every c — a partial stratum over all of $\mathbb{C}$, which is the opposite of membership. Membership is $M \equiv 0$, i.e. $\text{rank } 0$. The two must be reported separately (Proposition 2), and conflating them turns the strongest possible existence statement into a false membership claim.*

7.8 Measurements

The route of §§7.4–7.6 is implemented as the `--prolong` option of `joca-thomas.sage`. What it shows:

- *Lemma 1 holds on the nose.* Reducing P modulo the generic cell lands in exactly $\mathbb{Q}[c][x, y, z, r, \Psi, \Psi']$ — 228 terms, degree 9, 0.1 s. No v , no Ψ'' , no jets: six non-constant generators, as predicted.
- *The differential engine is not needed.* The explicit derivation (7) agrees with BLAD’s `differential_prem` on every cross-check, and is about **2000**× faster in the prolongation loop (0.1 s versus 200 s for one order). Building the derivation table costs a handful of single-jet reductions (milliseconds); after that the engine is never called again.
- *On the generic cell the whole differential side is sub-second.* Prolonging to $k = 2$ costs **0.3–0.7 s** at a few hundred MB, against a 6-hour timeout for the Thomas re-completion of the same cell. The claim of §7.3 is discharged *there*: the wall was the re-completion, not the reduction.
- *But the degenerate cells are not yet reachable — for a reason that is about the input, not the method.* Cells 26/27, which carry the $E = -1/2$ and $E = -1/8$ strata, are refused by Criterion 3: the chain `joca-thomas.sage` hands them is the re-specialized raw ansatz, not the completed Thomas cell, and Ψ'' escapes the ring (Remark 4). The fix is to feed the cells’ own equations from the Thomas log (`parse_diff=True`), which are already recorded there; until that is wired, **no existential claim about the strata-carrying cells is supported**, and the two earlier “results” this note might have reported for them were artifacts.
- *Coefficients stay small, ≤ 19 bits at $k = 3$ — the same verdict the staging diagnosis reached by measurement [8], now in the fixed-chain setting too.*
- *The polynomials do grow, and the wall moves to the commutative side.* Unpruned, the generators go $228 \rightarrow \sim 10^3 \rightarrow \sim 10^4 \rightarrow \sim 5 \cdot 10^4$ terms with degree +5 per order (the degree step is the cleared denominator $m = r(a_0 + a_1 v)$ accumulating; stripping the exact unit factors recovers roughly a third of it). On the generic cell the 2×2 minors of the $k \leq 2$ matrix reach $3.5 \cdot 10^5$ terms at degree 32, and the symbolic elimination over the eleven constants does not complete — the same parametric-elimination cliff as [6], now met on a much smaller problem.
- *The pointwise probe decides membership of $\mathcal{E}$ in 0.4 s.* Fixing c^* and asking whether the specialized minors have a common zero on the quadric is a Gröbner computation in *four* variables. It reports the recorded $E = -1/2$ membership point as lying in $\mathcal{E}$ — with the minors vanishing *identically*, i.e. correctly diagnosed as a *membership* point — and reports 0/5 random constant points in $\mathcal{E}$, so the locus is a proper subvariety and the test discriminates. This is the Nullstellensatz probe of [6, Open problem 1], designed there and never executed; the rank formulation makes it a few lines.

The honest summary. **On a genuinely passive cell the fixed-chain route removes the differential wall entirely**, and moves the problem onto the commutative side, where the pointwise probe is cheap (0.4 s) and the symbolic elimination is still the open cliff. Two things are therefore true at once, and neither should be overstated:

- the differential obstruction that made per-cell staging fail — the re-completion — is *gone*, replaced by table-lookup differentiation in a fixed ring; and

- the route has not yet been *run* on the cells that matter, because `joca-thomas.sage` does not currently feed them a passive chain (Remark 4). That is a wiring defect with a known fix, not a new wall — but it is unfixed, and the strata claims stay open until it is.

7.9 What it costs

Three honest debits, none fatal, all worth stating.

- *It uses the same reduction engine.* If $\partial^2 \rho_j$ modulo cell 1 happens to blow up to 10^5 terms anyway, the split-avoidance bought nothing. The structural argument above says it should not; only measurement settles it. Fortunately the measurement is cheap — ρ_j is already computed for all 29 cells; prolonging it two or three times and counting terms is an afternoon, not a task.
- *The final elimination is the known cliff, at a much smaller scale.* The global bounded-prolongation attempt did not complete even over $\text{GF}(32003)$ at $N = 2$ — but that ideal carried every jet of Ψ, v, r up to order N as an indeterminate, plus the whole ansatz’s prolongation generators. Here the ansatz has been *solved away by the reduction* and six non-constant variables remain; on degenerate cells the constants are partly specialized already (cell 26 has $a_0 = a_1 = 0$, so nine constants, not eleven). A different-sized problem, not the same cliff — but not a guarantee either.
- *It returns $\overline{\mathcal{E}}_j$, not disjoint strata.* That is the price of not splitting, and it is exactly the trade §2.4 names: the single-ideal route returns the Zariski closure, a superset that never loses a component. For the question actually on the table — *does this cell carry a partial stratum, and at which constants* — closure is enough.

8 Open problems and next steps

1. **No a-priori closure order for $A \cup \{P\}$.** The ansatz bound $N_0(A) = 2$ does not bound the combined system; the constants-cutting conditions $D^k \rho$ enter at order $2 + k$ and the terminating k is unknown for hydrogen (heuristically ~ 4 – 6 ; unmeasured). The pointwise probe `prolong-probe.sage` is written to lower-bound it empirically but has not yet produced data [6]. An a-priori bound in general would bear on the effectivity gap of the Ritt problem [16, 17] — do not expect one; prolong-until-stable is the algorithm.
2. **The parametric elimination cliff.** Direct $\text{III}_{\exists}$ over the eleven hydrogen constants as ring variables did not complete even over $\text{GF}(32003)$ at order 2. The recommended path is per-cell staging over the 29 already-computed Thomas cells, with the cheap membership filter ($\mathcal{M} \subseteq \mathcal{E}$) applied first; syzygy-based consistency extraction is the principled single-ideal fallback [6]. Until one of these lands, the corrected algorithm’s Phase $\text{III}_{\exists}$ is validated only on toy examples (§5.3).

Update (§7): per-cell staging in its obvious form — re-run differential Thomas on $T_j \cup \{P\}$ — itself fails on five of the 29 cells [8], and the failure is not coefficient swell but intra-step live-set swell inside the re-completion. The fixed-chain prolongation of §7.4 is the proposed way past it: it deletes the re-completion, keeps the reduction (which is measurably cheap on all 29 cells), and reduces the whole per-cell $\exists$ question to a commutative D -closure computation in six non-constant variables. **Unmeasured:** whether $\partial^k \rho_j$ stays small under (7). That is the next experiment, and it is cheap.

3. **The “used inequations” completeness argument.** Phase II splits only on tracked divisors. The argument that this suffices — untracked initials/separants do not affect the reduction identities, and the leaf’s own solved-form initials are always used — should be written out against Bächler et al.’s simple-system conditions, in particular the squarefreeness splits (discriminants), which the tracking as described captures only when a gcd computation actually divides by them. Until then, treat Phases I–II as “Thomas with a lazy strategy, conjecturally split-minimal,” not as a proven refinement.
4. **Real witnesses.** $\text{III}_{\exists}$ certifies complex existence. For quantum-mechanical conclusions the real (and normalizability) questions are downstream filters on the resulting varieties, as in the paper’s treatment; a systematic real-existential layer (real QE / CAD) is unexplored territory for this problem class.
5. **Verification against the recorded runs.** The concrete check that would close the loop: run $\text{III}_{\exists}/\text{III}_{\forall}$ per cell on the 29 hydrogen cells and confirm (i) the membership cells’ union reproduces `joca.sage`’s five primes, (ii) the partial strata include the $E = -1/2$ and $E = -1/8$ cells with their recorded primes, and (iii) nothing else appears. The classifier of `joca-rg-combined.sage` already implements the $\text{III}_{\forall}$ test and validates on the toy examples; the hydrogen combined RG run itself does not terminate, which is precisely why the per-cell route matters.
6. **GF(32003) vs $\mathbb{Q}$.** Any stratum destined for the paper needs a characteristic-zero confirmation of the modular computations.

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Researched and written by Claude (Fable 5) on behalf of Brent Baccala, task 463 of the project task runner; the worked examples were verified by hand against the recorded outputs of `joca.sage`, `joca-rg.sage`, `joca-rg-combined.sage`, and the open-maple differential Thomas run.

/claude-fable-5